HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Automated diagnostic data collection for ~150 industrial control systems using **Python scripting**, reducing manual effort by 80% and enabling faster root cause analysis.
- Developed and maintained **Java Spring Boot** REST APIs to integrate disparate operational data sources, supporting seamless data flow between enterprise systems and the automation platform.
- Managed production incidents for enterprise applications, performing **root cause analysis** and implementing fixes within strict SLA targets, ensuring high availability for critical business operations.
- Designed and implemented **PLC and SCADA** logic for a new pharmaceutical manufacturing line, ensuring compliance with stringent industry standards and improving process control accuracy by 15%.
- Generated comprehensive **technical documentation** for automation system designs and control panel hardware specifications, facilitating client understanding and internal knowledge transfer.
- Collaborated with customer engineering teams to define requirements and execute site acceptance testing for a new **manufacturing execution system (MES)** implementation, ensuring successful deployment.
- Containerised core backend services using **Docker** and deployed them to an AWS EKS cluster, streamlining the deployment process and improving application scalability.

SKILLS

Industrial Automation – PLCs, SCADA, Instrumentation, Control Systems, MES, Industrial Control Systems

Programming Languages – Java, Python, SQL, Shell Scripting

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn

Cloud & DevOps – AWS, EC2, S3, Lambda, CloudWatch, Docker, Kubernetes, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

Tools & Technologies – Splunk, Dynatrace, Jira, Confluence, Linux

PROJECTS

Industrial Control System Automation Suite

- Developed a **Python**-based automation suite to manage and monitor a fleet of 200+ industrial control systems, integrating with **SCADA** interfaces for real-time data acquisition.
- Implemented automated health checks and diagnostic data collection scripts, reducing system downtime by an average of 20% and providing actionable insights for proactive maintenance.
- Engineered **REST APIs** using Flask to expose system status and performance metrics, enabling integration with enterprise monitoring dashboards and facilitating faster incident response.

Pharmaceutical Process Control System Upgrade

- Designed and programmed **PLC** logic for a critical upstream processing unit in a pharmaceutical manufacturing environment, adhering to strict validation and GAMP guidelines.
- Developed custom **SCADA** screens and alarm management configurations to provide operators with clear visibility and control over the automated process, improving operational efficiency.
- Collaborated with client stakeholders throughout the UAT phase, incorporating feedback to refine control strategies and ensure seamless integration with existing plant infrastructure.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)